

Lecture 2

The fundamental principle of counting

Suppose we have k tasks to do and the i -th task can be done in n_i ways.

In how many ways can you do the k tasks?

$$\text{ANSWER} = \prod_{i=1}^k n_i = n_1 \cdot n_2 \cdot \dots \cdot n_k.$$

We assume that every task
DO NOT affect any other task.

WE FOCUS ON 3 DIFFERENT SITUATIONS:

- ① We have a deck of cards and we want to put them down on the table one by one. In how many different ways I can do that?

$$\text{ANS} = 54 \cdot 53 \cdot \dots \cdot 1 = 54! \quad (n! = n \cdot (n-1) \cdot \dots \cdot 2 \cdot 1)$$

Def: Given a number $n \in \mathbb{N}$, the FACTORIAL of n is denoted by $n!$ and is equal to $n \cdot (n-1) \cdot \dots \cdot 2 \cdot 1$. [Convention: $0! = 1$]

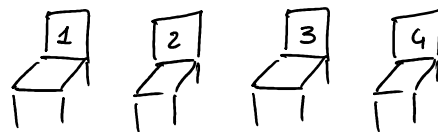
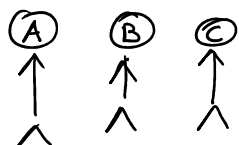
Remark: $n!$ counts the possible ways of reordering n distinct objects.

Remark: $n!$ is counting the # of permutations of a set of n numbers.

Example: $[3] = \{1, 2, 3\}$ $S_3 = \{123, 132, 213, 231, 312, 321\}$

Note that $|S_3| = 3! = 6$. In general, $|S_n| = n!$.

- ② We want to seat k guests on n chairs ($k \leq n$)



In how many ways I can do that?

$$\text{ANS} = n (n-1) \dots (n-k+1) = \frac{n!}{(n-k)!}$$

③ We want to "Select a SET of k chairs out of n chairs" ($k \leq n$)



In how many ways I can do that?

$$\text{ANS} = x$$

Consider the previous task: "Setting k guests on n chairs" Example ② $\left[\# \text{ ways} = \frac{n!}{(n-k)!} \right]$

This task can be also done in the following way:

① - We select k chairs among the n chairs $\left[\# \text{ ways} = x \right]$

② - We seat the k guests on the k chairs selected at the previous step. $\left[\# \text{ ways} = k! \right]$

Hence:

$$\frac{n!}{(n-k)!} = x \cdot k! \implies x = \frac{n!}{(n-k)! \cdot k!}$$

Definition: The BINOMIAL COEFFICIENT $\binom{n}{k}$ is defined as

$$\binom{n}{k} := \frac{n!}{(n-k)! k!} \quad (k \leq n)$$

Remark: The terminology is a consequence of this fact:

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} \quad \left[\text{Binomial theorem} \right]$$

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Exercise: Prove it!

Coming back to the example ③ of Lecture 1:

③ Experiment is "We toss a coin n times", where $n \in \mathbb{N}$, then

$$\Omega = \{ ? \} \quad \text{but} \quad |\Omega| = 2^n$$

and what is a "natural" prob. measure?

$$P(\omega) = \frac{1}{|\Omega|} = \frac{1}{2^n} = 2^{-n}, \quad \forall \omega \in \Omega.$$

Now we fix $k \in \mathbb{N}$, $0 \leq k \leq n$.

$$A = \{ \text{Tossing the coin } n \text{ times we get } k \overset{\text{"H"}}{\uparrow} \text{ heads} \}$$

$$P(A) = ? = \frac{|A|}{|\Omega|} \quad \text{but} \quad |A| \text{ is a bit a mess!}$$

$$\Rightarrow P(A) = 2^{-n} \cdot |A| = \binom{n}{k} \cdot 2^{-n} = \frac{n!}{(n-k)! k!} \cdot 2^{-n}$$

↳ We just need to SELECT the set of the k indexes among the n coins where we want to put H

$$\Rightarrow \# \text{ ways} = \binom{n}{k}$$

Example: THE BIRTHDAY PROBLEM.

people in the class = n

Q: What is the probability that at least 2 people are born in the same day of the year?

- We assume that there are 365 days a year.
- We also assume that every day of the year there is the same chance to be born.

$P(\underbrace{\text{At least 2 birthdays are on the same day}}_A)$

$$\begin{aligned} P(A) &= 1 - P(A^c) = 1 - P(\underbrace{\text{All birthdays are different}}_{\Omega \setminus A}) = 1 - \frac{365!}{365^n (365-n)!} \\ &= \frac{365 \cdot 364 \cdot \dots \cdot (365-n+1)}{365^n} \end{aligned}$$

$$\text{When } n=40 \implies P(A) = 0.89$$

Operations with events

Recall that an EVENT is a subset of the sample space Ω .

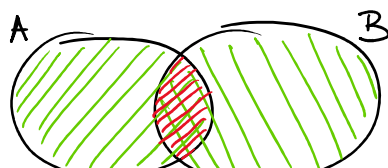
Def: The COMPLEMENT of an event A , denoted by A^c , and is equal to $A^c = \Omega \setminus A$.

Note that $P(A^c) = 1 - P(A)$, indeed $P(A^c) = P(\Omega \setminus A) = \underbrace{P(\Omega)}_1 - P(A) = 1 - P(A)$.

Def: The UNION of two events A & B is denoted by $A \cup B$, and it contains all $w \in \Omega$ s.t. $w \in A$ or $w \in B$.

Def: The INTERSECTION of two events A & B is denoted by $A \cap B$, and it contains all $w \in \Omega$ s.t. $w \in A$ and $w \in B$.

Example:



$A \cup B$ 

$A \cap B$ 



$$A \cap B \quad \text{shaded}$$

Some computations: Recall that $P(A) = \sum_{\omega \in A} P(\omega)$

$$P(A \cup B) = \sum_{\omega \in A \cup B} P(\omega) = \sum_{\omega \in A} P(\omega) + \sum_{\omega \in B} P(\omega) - \sum_{\omega \in A \cap B} P(\omega) = P(A) + P(B) - \underbrace{P(A \cap B)}_{\geq 0}$$

As a consequence: $P(A \cup B) \leq P(A) + P(B)$ ["Union bound"]

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i)$$

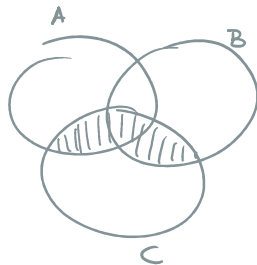
What about an exact formula for $P(A \cup B \cup C)$?

$$P(A \cup B \cup C) = P((A \cup B) \cup C) = \underbrace{P(A \cup B)}_{P(A) + P(B) - P(A \cap B)} + P(C) - \underbrace{P((A \cup B) \cap C)}_{(*)}$$

Let's compute

$$(*) = P((A \cup B) \cap C) = P((A \cap C) \cup (B \cap C)) = P(A \cap C) + P(B \cap C) - \underbrace{P((A \cap C) \cap (B \cap C))}_{A \cap B \cap C}$$

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$



We get

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

From here, we can guess the following general formula:

From here, we can guess the following general formula:

$$P(A_1 \cup \dots \cup A_n) = \sum_{i=1}^n P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) + \dots + (-1)^{n+1} P(A_1 \cap \dots \cap A_n)$$

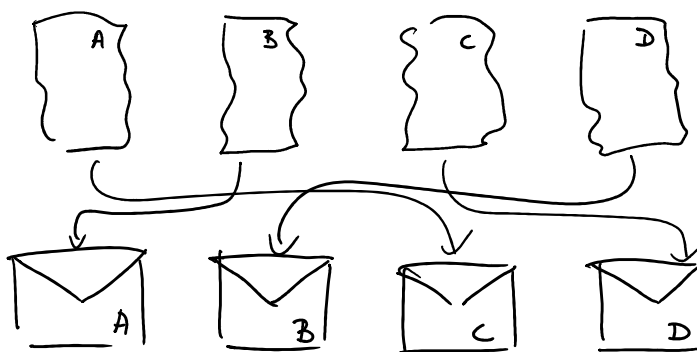
"INCLUSION-EXCLUSION FORMULA"

Exercise: Prove the formula by induction

[Hint: copy what we did to go from 2 events to 3 event for going from n event to (n+1) events]

Example: [THE SECRETARY PROBLEM]

We have (n) letters & n envelopes:



We put each letter in an envelope unif. at random.

Q: What is the probability that the letters are all in the WRONG envelope?